

Reviewer Pack — Minimal Rank of Cluster Algebras and Communication Complexit...

Entry 0722c204b678 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Cluster Algebras and Communication Complexity Rank

Entry ID: 0722c204b678 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-01 20:36:22 UTC

1. Conjecture

Field A (mathematical branch): Cluster Algebras **Field B** (complexity object): Communication Complexity

Statement:

For every d -regular graph G , the minimal rank of its associated cluster algebra $c(G)$ is linearly correlated with its communication complexity rank $r(G)$, such that $\min_rank(c(G)) = \Theta(r(G))$.

Rationale (proposer's reasoning):

Cluster algebras encode combinatorial structures that can capture the complexity of certain discrete objects. The minimal rank of a cluster algebra might provide insight into the underlying complexity of a graph's communication, offering a novel way to relate algebraic invariants to complexity measures.

Taxonomy category: ClusterAlgebraCommunicationComplexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 5cb8cebb9daf8db9

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For each d -regular graph G , if the absolute difference between $\min_rank(c(G))$ and $r(G)$ is less than or equal to 1.5 standard deviations of their mean differences over all seeds, then we accept the conjecture.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 10 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - cluster algebras AND communication complexity rank - minimal rank cluster algebra AND d-regular graph communication complexity - $\theta(\text{minimal rank } c(G)) = \theta(r(G))$ cluster algebra communication complexity

Top relevant hits considered: - [http://arxiv.org/abs/1410.5374v2] Cluster algebras of infinite rank as colimits - [http://arxiv.org/abs/1101.4803v2] Lifting defects for nonstable K_0 -theory of exchange rings and C^* -algebras - [http://arxiv.org/abs/1008.1829v2] On cluster variables of rank two acyclic cluster algebras - [http://arxiv.org/abs/0810.4836v3] Simplicial complexes and minimal free resolution of monomial algebras - [http://arxiv.org/abs/1102.2932v2] Applications of Monotone Rank to Complexity Theory - [http://arxiv.org/abs/0901.0512v4] Expected Performance of the ATLAS Experiment - Detector, Trigger and Physics - [http://arxiv.org/abs/1411.4413v2] Observation of the rare $B_s^0 \rightarrow \mu^+ \mu^-$ decay from the combined analysis of CMS and LHCb data - [http://arxiv.org/abs/1406.6311v4] The Physics of the B Factories

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction
import itertools

# Gaussian elimination to reduce matrix to row echelon form
def gaussian_elimination(matrix):
    n = len(matrix)
    for i in range(n):
        # Find max element in current column
        max_row = max(range(i, n), key=lambda r: abs(matrix[r][i]))
        if matrix[max_row][i] == 0:
            raise ValueError("Singular matrix")

        # Swap rows
        matrix[i], matrix[max_row] = matrix[max_row], matrix[i]

        # Eliminate current column elements below pivot
        for j in range(i + 1, n):
            factor = Fraction(matrix[j][i], matrix[i][i])
            for k in range(n):
                matrix[j][k] -= factor * matrix[i][k]

    return matrix
```

```

# Compute the rank of a matrix using Gaussian elimination
def min_rank(matrix):
    try:
        reduced_matrix = gaussian_elimination(matrix)
        rank = sum(1 for row in reduced_matrix if any(row))
        return rank
    except ValueError as e:
        return None

# Generate a random d-regular graph with n vertices
def generate_d_regular_graph(n, d):
    if (n * d) % 2 != 0:
        raise ValueError("d must be even for regular graphs")

    edges = set()
    while len(edges) < n * d // 2:
        u, v = random.sample(range(n), 2)
        if u > v:
            u, v = v, u
        if (u, v) not in edges and (v, u) not in edges:
            edges.add((u, v))

    adjacency_matrix = [[0] * n for _ in range(n)]
    for u, v in edges:
        adjacency_matrix[u][v] = 1
        adjacency_matrix[v][u] = 1

    return adjacency_matrix

# Compute the communication complexity rank of a graph
def communication_complexity_rank(graph):
    n = len(graph)
    max_flow = 0
    for source in range(n):
        visited = [False] * n
        flow = 0
        queue = [source]
        while queue:
            u = queue.pop(0)
            if visited[u]:
                continue
            visited[u] = True
            for v in range(n):
                if not visited[v] and graph[u][v] > 0:
                    queue.append(v)
                    flow += min(graph[source][u], graph[u][v])
        max_flow = max(max_flow, flow)

    return max_flow

# Run a single trial with the given seed
def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    results = []

```

```

for n in n_values:
    d = 2 * random.randint(1, n // 2)
    graph = generate_d_regular_graph(n, d)
    cluster_algebra = graph # Simplified representation for testing
    min_rank_cG = min_rank(cluster_algebra)
    r_G = communication_complexity_rank(graph)

    if min_rank_cG is None:
        continue

    results.append({
        "n": n,
        "min_rank_cG": min_rank_cG,
        "r_G": r_G
    })

if not results:
    return {
        "metric_name": "min_rank_cG vs r_G",
        "metric_value": None,
        "instances_tested": 0,
        "n_max": 0,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

min_rank_values = [result["min_rank_cG"] for result in results]
r_G_values = [result["r_G"] for result in results]
mean_min_rank = sum(min_rank_values) / len(min_rank_values)
mean_r_G = sum(r_G_values) / len(r_G_values)
std_dev = math.sqrt(sum((x - mean_min_rank)**2 for x in min_rank_values) / len(min_rank_values))

correlation = sum((min_rank_values[i] - mean_min_rank) * (r_G_values[i] - mean_r_G) for i in range(len(min_rank_values))) / (len(min_rank_values) * std_dev)

return {
    "metric_name": "min_rank_cG vs r_G",
    "metric_value": correlation,
    "instances_tested": len(results),
    "n_max": max(result["n"] for result in results),
    "conjecture_holds": abs(correlation) >= 1.5 * std_dev,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2**i - 1 for i in range(2, 31)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, ...{result}}}")
        results.append(result)

    if all("conjecture_holds" not in r or r["conjecture_holds"] for r in results):
        mean_value = sum(r["metric_value"] for r in results) / len(results)

```

```

std_dev = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
support_fraction = sum(1 for r in results if "conjecture_holds" not in r or r["conjecture_holds"] == "yes") / len(results)
print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
elif any("counterexample" in r and r["counterexample"] for r in results):
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if "counterexample" in result)
    print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means that the pre-registered support condition could not be evaluated. | next: Re-run the test with a longer timeout or an alternative method to ensure it completes and produces results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14903
2	preregistration	ollama_remote	glm4:latest	0	0	9189
3	novelty	ollama_remote	glm4:latest	0	0	8754
4	novelty	ollama_remote	glm4:latest	0	0	11484
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16913
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13665
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12528
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15897
9	judge	ollama_remote	glm4:latest	0	0	9153

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 112486 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in <research/audit/0722c204b678.jsonl>)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/0722c204b678.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/0722c204b678.tar.gz` (if generated)